

Project-01: Home Penetration Testing Lab:

Built an isolated **virtual penetration testing lab** using **VirtualBox**, simulating real-world attack and defense scenarios using **Kali Linux** and vulnerable machines (**Metasploitable**, **DVWA**). Configured internal networks, firewalls, and tested exploits using tools like **Nmap**, **Metasploit**, and **Burp Suite**.

Project Setup

This guide will teach you how to set up your own private lab. To do this, we'll need three things:

- Virtualization software
- Attacking Machine
- Target Machine

Virtualization software allows one physical computer to run multiple virtual machines (VMs). A virtual machine functions like a separate computer, complete with its own operating system.

VirtualBox is a popular virtualization software. VMware is another alternative.

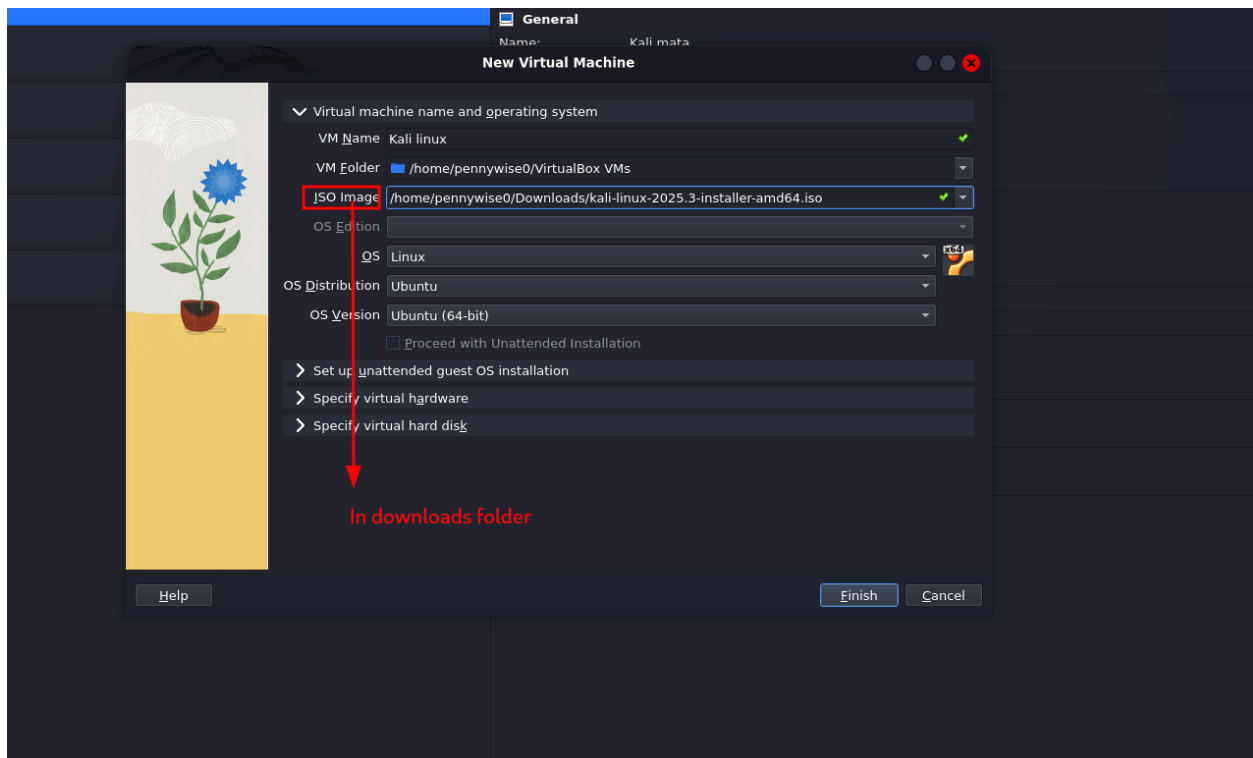
To practice hacking, you need two machines—an attacking machine and a target machine.

The downloads required for this setup are quite large.

- [Download VirtualBox](#) (download the extension pack as well)
 - [Download Kali](#) (64-bit Virtualbox image or .iso file)
 - [Download Mr Robot vulnerable machine](#)
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How to Install Kali Linux

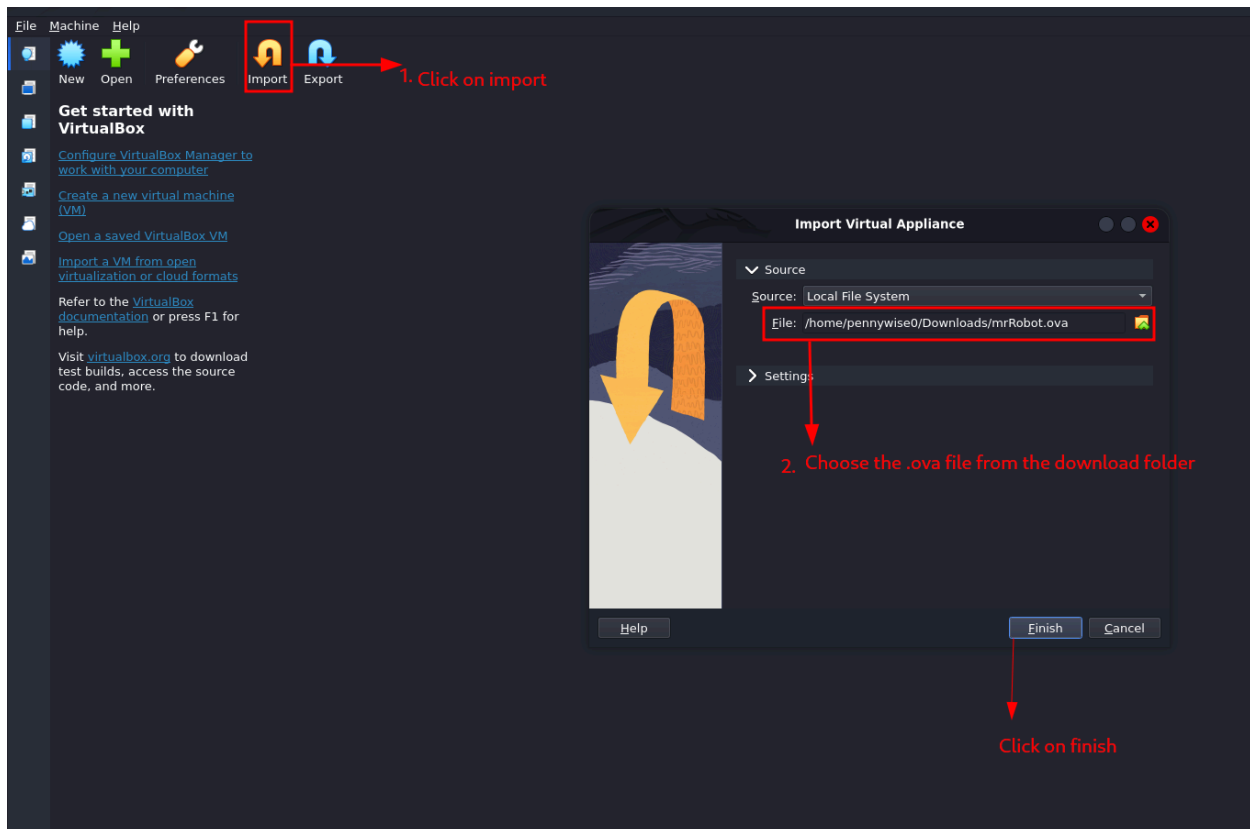
Now, let's install the Kali Linux attacking machine. Then click the green "Add" icon on the VirtualBox interface and point to the .iso file.



All the default settings will be applied, and you should have the attacking machine installed.

How to Install a Target VM

Now let's install the target. Double-click on the downloaded `mrRobot.ova` file. Use the default settings and click "Finish".

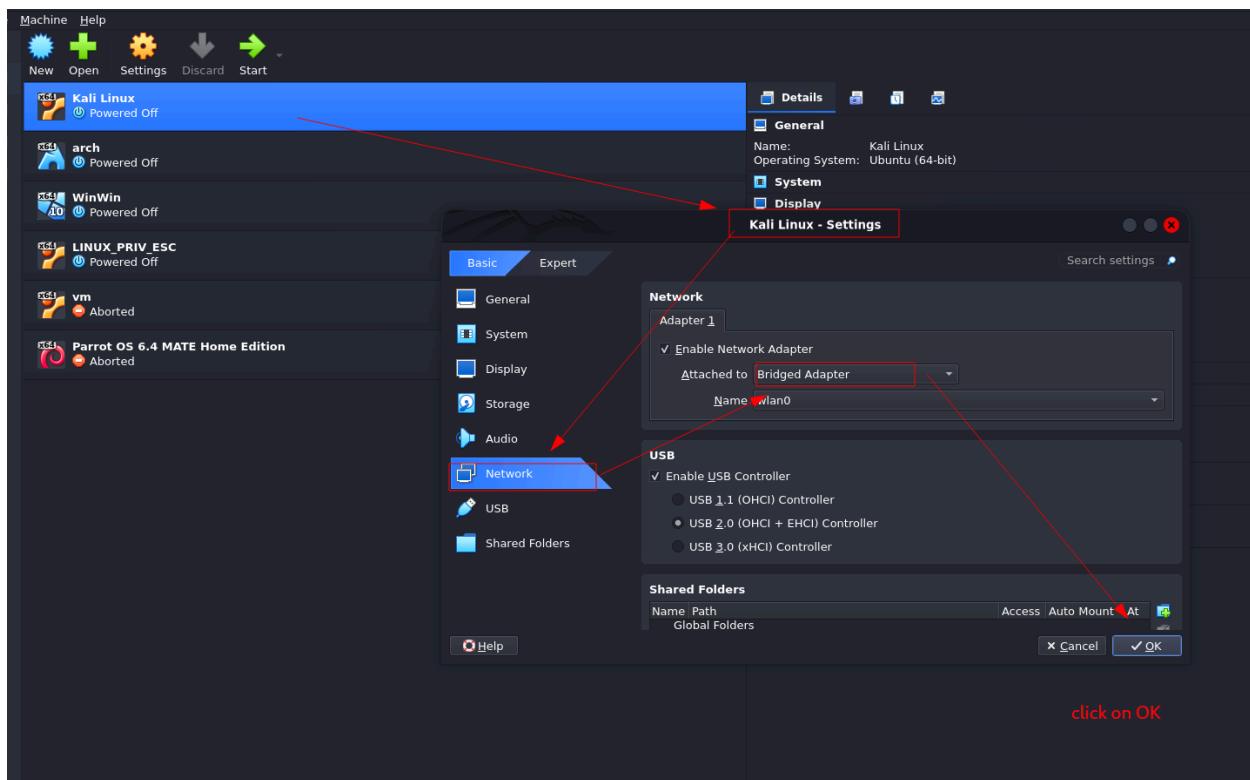


Update Networking Settings

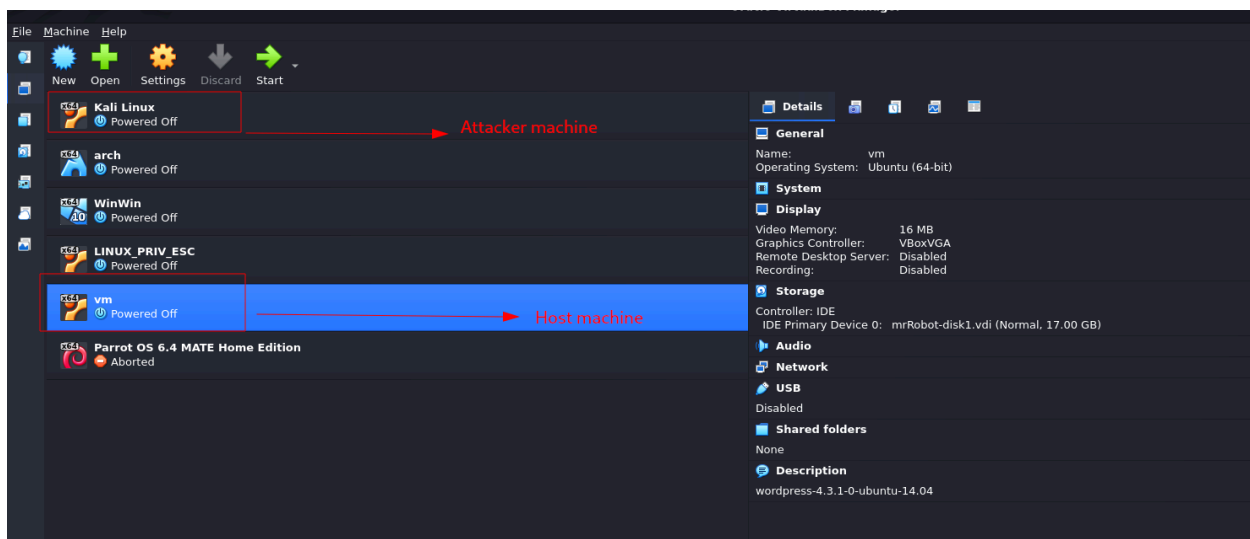
There are many ways to set up a network in VirtualBox. But in this case we want to isolate our lab from the public internet. The best way to do this is to set up a host-only network.

In a host-only network, the VMs can communicate with each other but not with the public internet. Let's set it up.

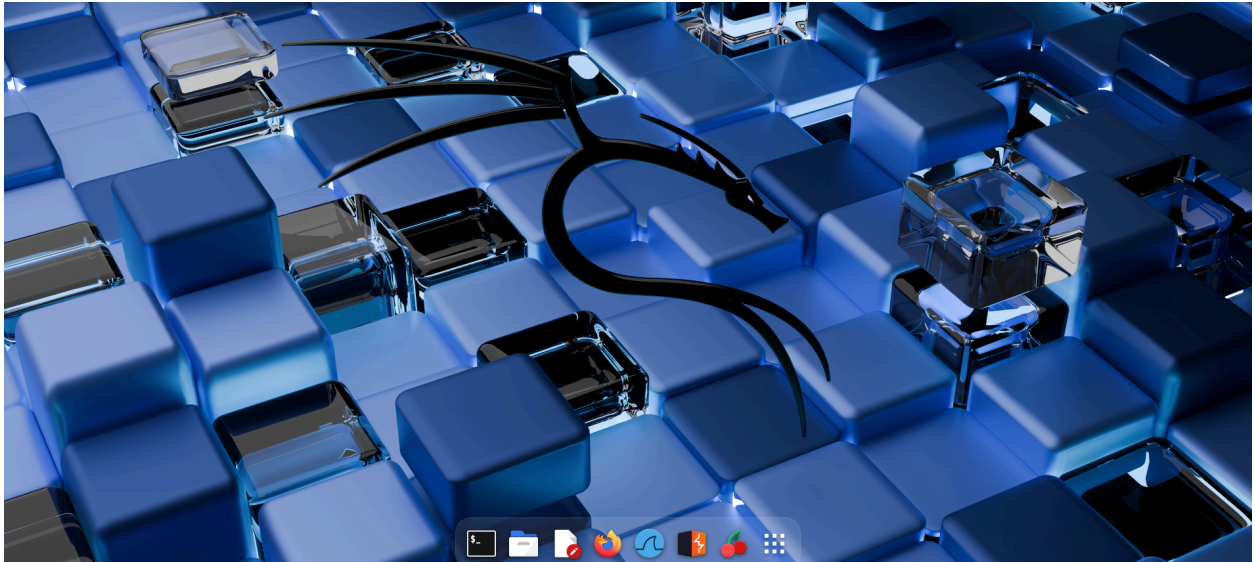
In the VirtualBox interface, click on "Tools" and click "Host-only Networks". Then click "Create". It will automatically create a host-only network with an IP range.



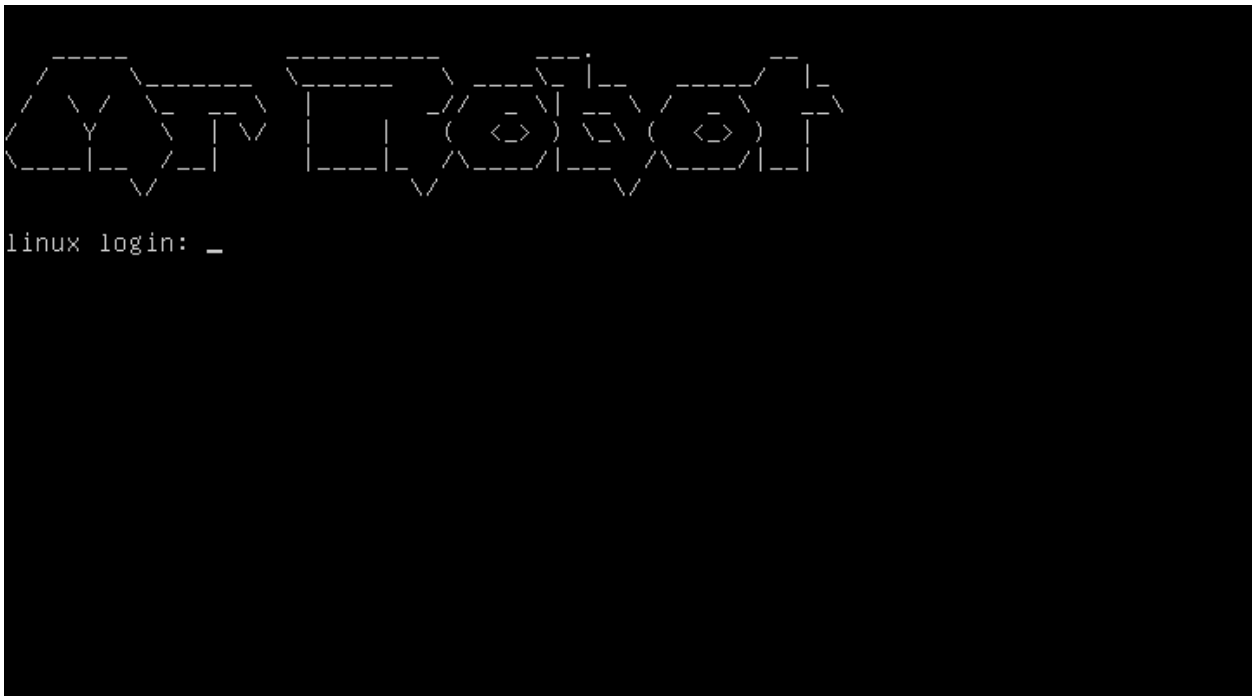
We are ready to boot the machines.



Kali Linux UI is shown below.



Mr.Robot box UI is shown below.



Now let's find the IP addresses of these machines.

```
penny@penny: ~  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:b8:01:e7 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.1.23/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0  
        valid_lft 85948sec preferred_lft 85948sec  
    inet6 2401:4900:1ca3:4f82:5fff:9d85:869b:ddd6/64 scope global temporary dynamic  
        valid_lft 85950sec preferred_lft 85471sec  
    inet6 2401:4900:1ca3:4f82:a00:27ff:feb8:1e7/64 scope global dynamic mngtmpadr noprefixroute  
        valid_lft 85950sec preferred_lft 85950sec  
    inet6 fe80::a00:27ff:feb8:1e7/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
(penny@penny)~  
$
```

Attacker machine IP and MAC address.

```
$ nmap 192.168.1.24  
Starting Nmap 7.95 ( https://nmap.org ) at   
Nmap scan report for 192.168.1.24  
Host is up (0.00041s latency).  
Not shown: 997 filtered tcp ports (no-response)  
PORT      STATE SERVICE  
22/tcp    closed ssh  
80/tcp    open  http  
443/tcp   open  https  
MAC Address: 08:00:27:A9:25:C9 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)  
Nmap done: 1 IP address (1 host up) scanned in 5.51 seconds
```

Host machine IP and MAC address.

Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts

11 Captured ARP Req/Rep packets, from 4 hosts. Total size: 552

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.1.1	5c:8c:30:aa:e5:c0	8	390	Taicang T&W Electronics
192.168.1.23	08:00:27:b8:01:e7	1	60	PCS Systemtechnik GmbH
192.168.1.24	08:00:27:a9:25:c9	1	42	PCS Systemtechnik GmbH
192.168.1.240	5c:35:48:80:df:5c	1	60	Aditya Infotech Ltd.

A quick comparison of the IP and MAC addresses of both machines.

Conclusion

We successfully set up your hacking lab using VirtualBox. This lab gives you the flexibility to practice ethical hacking in a controlled, isolated environment.